



Non Functional Requirements

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Core Non Functional Requirements

The core non functional requirements of the system are detailed as follows:

- **Security:** All sensitive user data should be encrypted using AES-256 encryption, and the system should enforce multi-factor authentication for administrative accounts.
- **Reliability:** The system should achieve 99.9% uptime and recover from critical failures within 5 minutes.
- **Performance:** The system should respond to user requests within 2 seconds for 95% of requests under normal operating conditions and support at least 60 concurrent users.
- **Availability:** The system should be available 24/7, excluding scheduled maintenance periods not exceeding 2 hours per month.
- **Maintainability:** New features or bug fixes should be deployable within 2 hours, and the codebase should maintain at least 80% automated test coverage.

Constraints

Open-Source Requirement

All components and third-party libraries must be open-source. This avoids licensing costs, vendor lock-in, and ensures the solution remains maintainable beyond the project lifecycle.

Deployment Requirement

The system must be fully containerised using Docker. All services must be orchestrated through Docker Compose and deployable via a single `docker compose up` command with no manual configuration steps. GitHub Actions is used for CI/CD to validate builds and run tests before any merge to the main branch. The production environment must enforce HTTPS and role-based access control.

Progressive Web App Requirement

The system must be delivered as a PWA. This is a hard requirement driven by the field environment where rangers operate without guaranteed connectivity. The PWA must support offline data capture and synchronise with the central database once connection is restored.

Data Sensitivity

The system handles sensitive conservation data including poaching incident locations, patrol routes, and informant tip-offs. All data in transit must be encrypted via HTTPS, and sensitive map layers must be restricted by authenticated user role.

Budget Constraint

The total project budget provided by EPI-USE is capped at R5000. All cloud services, third-party integrations, and tooling must be operated within this limit.

Technology Requirements

The technology requirements for Savanna Sentinel encompass the architectural and delivery requirements that ensure the system supports a reliable, scalable, performant, and secure Progressive Web App (PWA). We have taken into consideration the following technology requirements while choosing the tech stack for Savanna Sentinel.

Technology Stack Overview

Frontend	React 19 (TypeScript)
UI Components	shadcn/ui + Tailwind CSS v4
Mapping	MapLibre GL
PWA / Offline	Workbox + Dexie.js (IndexedDB)
Backend API	Python FastAPI
Background Jobs	Celery + Redis
Database	PostgreSQL + PostGIS
Object Storage	MinIO
AI / ML	sckit-learn + GeoPandas + pandas + SHAP
Security	JWT (PyJWT, HS256)
Reverse Proxy	Caddy
Email	Resend SDK
DevOps / CI	Docker + Github Actions

Frontend

Chosen Technology: React with TypeScript

React was explicitly recommended by our client EPI-USE in the project proposal. Beyond the client's preference, it is the most suitable choice for the following reasons:

- **PWA support:** React integrates natively with Workbox service workers, satisfying the offline field capture requirement without an additional framework layer.
- **Type safety:** TypeScript catches type mismatches between geospatial API responses and map layer data at compile time, reducing the class of bugs most likely to occur.
- **MapLibre GL integration:** The `react-map-gl` binding for MapLibre GL supports the heatmap layer, time range slider, and role based layer visibility without custom WebGL code.
- **Component isolation:** The explainability panel, patrol comparison interface, and dashboard cards can be developed as independent components, supporting parallel development.

Alternative considered - Angular: Angular was listed in the proposal as an option. We decided against it, because the team's existing React/TypeScript proficiency would have required relearning Angular's dependency injection and zone based change detection, reducing development velocity without any functional benefit for this project.

UI Components & Styling

Chosen technology: shadcn/ui with Tailwind CSS

shadcn/ui is a React component library built on Radix UI primitives and styled with Tailwind CSS utility classes.

- **Ownership model:** shadcn components are copied directly into the project codebase rather than installed as a black-box dependency. This means every component (buttons, forms, modals, navigation) can be customised to match the Savanna Sentinel brand palette and typography without fighting an external design system.
- **Tailwind integration:** Tailwind's utility-first CSS classes provide precise layout and spacing control across the dashboard, map interface, and mobile PWA forms without writing custom CSS files.
- **Lucide icons:** shadcn ships with Lucide as its default icon set, which is already specified in the Brand Style guide. No additional icon library is required.
- **Accessibility:** shadcn components are built on Radix UI, which provides ARIA roles, keyboard navigation, and focus management out of the box, directly supporting the WCAG 2.1 AA accessibility requirements.
- **Radix UI primitives:** Complex interactive components such as modals, dropdowns, sliders (time range), and tooltips (explainability panel) are handled by Radix UI's headless primitives, ensuring correct behaviour without custom implementation.

Alternative considered - Bootstrap: Bootstrap was considered due to team familiarity. It was not selected because it ships its own opinionated CSS that conflicts with Tailwind utility classes, requires a separate react-bootstrap wrapper package to function properly in React, and makes it significantly harder to apply a custom brand palette without overriding its defaults. shadcn + Tailwind provides the same component coverage with full control over styling and no class-name conflicts.

Mapping

Chosen technology: MapLibre GL with OpenStreetMap tiles

MapLibre GL renders map tiles and overlay layers using the device GPU via WebGL rather than browser side JavaScript loops, which is critical for the project.

- **Heatmap performance:** Risk heatmaps covering large reserves (thousands of grid cells) must render smoothly as the user drags the time slider. CPU based libraries such as Leaflet with canvas heatmap plugins stutter above approximately 5 000 points on mid range devices.
- **No licensing cost:** MapLibre GL is fully open source using free OpenStreetMap tiles.
- **Vector tile support:** MapLibre supports vector tile sources, enabling role based layer toggling at the client without additional server round trips. Sensitive layers can be managed by role based access control.

Alternative considered - Leaflet: Leaflet was evaluated due to its simplicity, large plugin ecosystem, and ease of integration with React. However, it was not selected because it relies primarily on CPU based rendering, which does not scale efficiently for large geospatial datasets. In performance testing scenarios relevant to this project (thousands of spatial points and dynamic heatmaps), Leaflet exhibits noticeable lag and reduced frame rates on mid range devices. Additionally, Leaflet lacks native vector tile support and GPU acceleration, requiring additional plugins and workarounds that increase complexity without achieving MapLibre GL's performance characteristics.

PWA / Offline

Chosen technology: Workbox (service workers) with Dexie.js as a typed IndexedDB wrapper for structured offline storage.

Offline capability is a first class requirement. Rangers operate in reserves with no mobile data coverage.

- Workbox precaches the application shell so the app loads without a network connection. The Background Sync API queues POST requests (incident reports, sightings) and replays them automatically when connectivity is restored.
- Dexie.js provides a clean, typed API over IndexedDB to store draft field reports locally on the device. Timestamp based conflict resolution is applied server side when the upload reaches the FastAPI backend.
- Photo uploads are queued as blob references in IndexedDB and uploaded to MinIO once online, avoiding data loss for large files.

Backend API

Chosen technology: Python FastAPI

FastAPI was recommended by the client and is the optimal choice for reasons beyond the client's preference.

- **OpenAPI auto generation:** FastAPI produces an `/openapi.json` specification automatically from Python type annotations, satisfying the architectural requirement for a versioned REST API with OpenAPI documentation without additional tooling.
- **Async I/O:** FastAPI runs on ASGI, enabling concurrent request handling for multiple rangers submitting field reports simultaneously without blocking.
- **Python ecosystem co-location:** The AI/ML modules (scikit-learn, GeoPandas, pandas, SHAP) run in the same Python environment as the API, eliminating the inter process serialisation overhead that would exist if the ML layer were written in a different language.
- **RBAC middleware:** FastAPI's dependency injection system is well suited to implementing role-checking middleware without cross cutting boilerplate.

Alternative considered - Node.js: Node.js was listed in the proposal. It was not selected because scikit-learn and GeoPandas do not have production equivalent JavaScript alternatives. Using Node.js would have required a separate Python microservice for ML, adding an inter service boundary and increasing operational complexity.

Background Processing

Chosen technology: Celery with Redis as the message broker

Risk score computation and heatmap generation are computationally expensive operations that must not block synchronous API requests.

- Redis as the message broker is lightweight, easy to containerise, and provides the pub/sub mechanism Celery requires. The same Redis instance is reused for API level response caching, reducing repeated PostGIS query load.
- Celery Beat allows periodic model retraining and heatmap refresh to be configured as cron style tasks, satisfying the architectural requirement for background jobs without manual intervention.
- Worker isolation ensures a long running scoring job cannot starve API workers, preserving the system's performance quality attribute.

Database

Chosen technology: PostgreSQL with PostGIS

The EPI-USE architectural requirements explicitly mandate PostGIS spatial indexing.

- **Spatial indexing:** PostGIS GiST and BRIN indexes support the spatial operations the risk engine depends on, `ST_Within`, `ST_DWithin`, and `ST_MakeGrid`, at the scale of millions of historical incident records.
- **ACID compliance:** Field report data is safety critical. PostgreSQL transactional guarantees ensure a partial sync from a ranger's device either fully commits or rolls back, preventing corrupt sighting records.
- **JSONB columns:** Heterogeneous tip off payloads and field report attachments are stored in JSONB columns, providing flexible schema within typed columns without sacrificing queryability.
- **Row-Level Security (RLS):** PostgreSQL RLS policies enforce role based data redaction at the database layer as a defence-in-depth measure supplementing application level RBAC.

Object Storage

Chosen technology: MinIO

CSV uploads and optional photo classifications require blob storage separate from the relational database.

- **API compatibility:** The application uses the standard boto3 Python SDK. If the project later migrates to AWS S3 or another provider, no code changes are required - only an environment variable update.
- **Self hosted within budget:** MinIO runs as a Docker container within the same Compose stack. There are no per-GB transfer fees during development.
- **Pre-signed URLs:** Allow the frontend to upload photos directly to MinIO without routing large files through the FastAPI backend, reducing server load.

AI / ML

Chosen technology: scikit-learn, GeoPandas, pandas, SHAP

The AI Risk Engine requires risk scoring, spatial feature engineering, explainability metrics, and general data processing. This combination maps directly to those four concerns.

- **scikit-learn:** Provides gradient boosting and ensemble classifiers (Random Forest, XGBoost-compatible pipelines) that are appropriate for tabular spatio temporal incident data. These models outperform deep learning on small structured datasets typical of reserve scale historical records.
- **GeoPandas:** Handles spatial feature engineering: converting incident coordinates into grid cells, computing kernel density estimates per cell, and joining patrol coverage data with incident locations.
- **pandas:** Used alongside GeoPandas for general data manipulation, cleaning, transformation, and preprocessing of tabular datasets before spatial operations and model training.
- **SHAP (SHapley Additive exPlanations):** Generates per-cell explainability metrics, detailing the contribution of each feature to a given risk score. This satisfies the functional requirement for an explainability panel that communicates model reasoning without overclaiming certainty.

Security

Chosen technology: PyJWT with HS256 symmetric signing

Authentication and RBAC require a stateless session mechanism compatible with both the web dashboard and PWA offline mode.

- **Stateless tokens:** JWT tokens are self contained, enabling FastAPI to validate requests without a database round trip per request, which is important for high frequency map tile and sync requests.
- **HS256 symmetric signing:** A single shared secret is used to both sign and verify tokens. This is appropriate for a monolithic deployment where all services share the same secret via environment variable.
- **Role claims in token payload:** The user's role (Ranger, Analyst, Community Liaison, Admin) is encoded in the JWT. FastAPI dependency injection reads this claim to enforce view and data restrictions without querying the database on each request.
- **Refresh token tracking:** Issued refresh tokens are stored in the database with their JTI, expiry, and revocation timestamp, enabling secure logout and token invalidation without sacrificing stateless access token validation.
- **HTTPS enforcement:** All endpoints are served over TLS via a Caddy reverse proxy with automatic certificate provisioning.

Reverse Proxy

Chosen technology: Caddy

All external traffic passes through Caddy before reaching the FastAPI backend.

- **Automatic TLS:** Caddy provisions and renews TLS certificates automatically, satisfying the requirement that no endpoint is accessible over unencrypted HTTP without manual certificate management.
- **Simple configuration:** Caddy's declarative Caddyfile syntax reduces operational overhead compared to alternatives such as Nginx, with no separate certbot process required.
- **Containerised deployment:** Caddy runs as a Docker container within the same Compose stack, keeping the full system deployable via a single `docker compose up` command.

DevOps / CI

Chosen technology: Docker with Docker Compose and GitHub Actions

The architectural requirements require containerised deployment and a CI pipeline with automated testing.

- **Docker Compose:** Defines the full local environment (FastAPI, Celery, Redis, PostgreSQL+PostGIS, MinIO, Caddy) as a single `docker compose up` command, ensuring all team members develop against identical dependencies.
- **GitHub Actions:** Runs on every push and pull request to main. The pipeline executes backend unit and integration tests (pytest), frontend unit tests (Vitest), SonarCloud static analysis, and Coveralls coverage upload. Merging to main is blocked if any stage fails.
- **Branch protection rules:** Enforce mandatory pull requests and passing CI checks before merging.
- **The main branch:** Always reflects a deployable state.

Transactional Email

Chosen technology: Resend (SMTP relay) with the resend Python SDK

A transactional email service is required to deliver password reset links generated by the password recovery flow.

- Resend provides a developer-focused SMTP relay with a free tier (100 emails/day) that operates within the project budget constraint.
- **SDK integration:** The resend Python package integrates directly into the FastAPI service layer. No additional worker or container is required; reset emails are dispatched inline during the forgot-password request handler.
- **Open standard:** Resend uses standard SMTP and a minimal REST SDK. Migrating to an alternative provider (Mailgun, SendGrid, AWS SES) requires only an environment variable and SDK swap with no architectural change.
- **No self-hosted mail server:** Running a self-hosted SMTP server (Postfix, Mailhog in production) within the Docker Compose stack was considered but rejected due to deliverability risks and the operational overhead of managing SPF/DKIM records within the project timeline.